

Iris: A Content Management System Supporting Typography and Accessibility

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Abstract

Higher education continues to face challenges in meeting students' accessibility needs for course content, including assignment instructions and reference materials. Furthermore, the complexity of accessibility concepts raises significant barriers to making necessary improvements. This work introduces Iris, a course content management system that focuses on presentation, interactivity, and accessibility improvements, with minimal effort required from authors. Iris provides accessibility, usability, and customization features, including font and spacing settings, reading rulers, and heading-based navigation, to readers throughout the platform. Additionally, using a large language model (LLM), Iris provides language-based accommodation through three strategies: hinting, simplifying, and explaining. This feature uses a simple one-click interface to uphold interactions with course staff. In line with universal design (UD) principles, Iris's features are designed for all students, but especially target students with disabilities and, for the LLM-based features, nonnative English speakers and students unfamiliar with the topic. Iris uses a custom document format and editor to enable its various features, prevent semantic HTML errors, and leave flexibility for future enhancements.

Iris can distribute text-based content for any course, but was initially evaluated in introductory computer science courses. We deployed the platform to display assignment instructions and conducted surveys in three quarters to evaluate students' experience. The most recent survey, conducted in Fall 2025, compares Iris with the previous system used in the course. The results reveal a significant preference for Iris. However, students with disabilities were underrepresented in the sample. In the future, we plan to conduct targeted research to address this issue.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**; • **Applied computing** → *Education*; • **Social and professional topics** → *People with disabilities*.

Keywords

Disability, Accessibility, Universal Design, Typography, Dyslexia, Computer Science Education, Large Language Models

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1 Introduction

As educators and institutions push toward including students of diverse backgrounds in computer science courses, one important consideration is the accessibility of course materials. Students with disabilities in higher education continue to report difficulties accessing course content. Furthermore, instructors may not be equipped to address these issues due to various factors, including a lack of institutional training and support [1].

One major source of accessibility recommendations is the Web Content Accessibility Guidelines (WCAG) published by the World Wide Web Consortium (W3C). The WCAG provides recommendations on topics including color contrast, text alternatives to images and videos, and keyboard-friendly navigation. Crucially, the authors note that WCAG conformance can impact users' experience regardless of disability [9]. A related consideration is typography, the appearance of written text. Typographical factors such as font style and size, line length and spacing, and choice of emphasis can impact readers' attention and the document's effectiveness [2].

At UC Santa Barbara, assignment instructions and other course materials are distributed using a variety of methods, including a learning management system (LMS) such as Canvas, a website created by the instructor, PDF files, etc. All of these media present their own accessibility and typographical challenges. With instructor-managed websites and PDF documents in particular, the burden of accessibility and typography falls primarily on the instructor.

2 Contributions

With Iris, we propose a content management system that provides accessibility, usability, and customization features to readers throughout the platform without any effort from the author except for creating the document. Iris then allows authors to easily address remaining accessibility and typography issues, such as alternative text for images and improper punctuation. By lowering the barrier to entry for many accessibility and usability features, we hope to increase the reader satisfaction and inclusivity of the course content. Additionally, to follow universal design (UD) principles on equitable use [4], Iris's features are designed to be useful to all users regardless of ability level. Iris has been used previously in introductory computer science courses, but it can display text-based content for any course.



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3 Implementation

Iris's web frontend includes various accessibility, usability, and customization features available across the platform. Its stylesheets and default settings follow typographic conventions, including appropriate font, spacing, and line length [2].

Readers can then customize Iris's style to their preferences and accessibility needs. Iris provides font options, including font size adjustment and typefaces that may benefit readers with dyslexia or low vision, including Comic Sans, OpenDyslexic, and Atkinson Hyperlegible. While research on the benefit of typefaces like OpenDyslexic is mixed [7], we provide several options so readers can select a font according to their preference. Character, word, line, and paragraph spacing options are also provided to increase comfort while reading, particularly for students with dyslexia [3]. Finally, light/dark theme options and color settings allow readers to further personalize their experience.

Iris also includes accessibility features extending beyond style changes. Reading rulers, screen overlays that follow the cursor to help readers keep their place in the text, are provided to improve comfort for some users with dyslexia [6]. Students can also perform large language model (LLM) queries on the selected text or surrounding context in order to find the meaning of a phrase, simplify a complex passage, or provide an appropriate hint to guide them in the right direction on an assignment. This feature targets students who are new to the topic or whose first language is not English. The LLM features use a simple one-click interface with predefined prompts, focusing on responding to basic queries while leaving more complex questions to course staff. This upholds student-instructor interaction, one of the core values of our system. Additionally, for security, students cannot send arbitrary prompts to the LLM as the prompt text is generated by Iris's server.

Finally, Iris implements various convenience features and rich content types to allow authors to focus on content rather than its presentation or internal implementation:

- Anchor links to quickly jump to a heading in the document
- Click-to-expand image previews
- Support for rich content including interactive questions, math expressions, TikZ drawings, sidenotes, block quotes, and stylized information boxes

To facilitate these features, Iris uses ProseMirror [5] as its document editor and adopts its model of storing documents. This model utilizes a tree of nodes (e.g., paragraph, block quote, figure, caption) represented by JSON objects that, where applicable, can have marks (e.g., bold, italic) applied to them. The JSON representation ensures documents can be processed (e.g., compiling LaTeX fragments to HTML code) and checked (e.g., for typographical errors) with relative ease. The rigid definition of the document model also allows the frontend to guarantee semantically correct HTML. While rigid, the document model can be extended to add additional content types, and its rendering on the frontend can be improved without any action by the author.

As WCAG conformance is a major indicator of a site's accessibility, we regularly use WebAIM's WAVE utility [10] and manual inspection to evaluate Iris's adherence to the WCAG. In combination with Iris's frontend features, we believe this ensures a better reading experience for students of all levels of ability.

4 Evaluation

Iris was deployed in two introductory computer science courses in several quarters to display programming assignment instructions. During this time, we conducted three surveys to measure Iris's efficacy.

In the first study, students were asked to choose between Iris and the previous system used for that course for one assignment. At the end of the assignment, we administered a short preliminary survey to gather students' opinions on the platform.

In the second study, Iris was used for the entire term. At the end of the term, we administered a preliminary survey to measure Iris's user experience. The survey first included the short version of the User Experience Questionnaire (UEQ-S) [8] to evaluate Iris on a standard scale. It then included demographic questions asking respondents whether English was their first language, whether they identified with any disability, and whether they typically use any assistive technologies. Finally, the survey assessed students' impressions of Iris in detail using several Likert-scale questions.

The third study compares Iris to the previous system used in the course. The class was given four assignments on Iris and three on the previous system before the survey was administered. The survey contained two sets of UEQ-S and Likert-scale questions, one for each platform, and also included the demographic questions from the previous survey.

5 Results and Discussion

The results of the two initial surveys indicated generally positive reception of Iris, but also highlighted room for improvement. These insights were used to improve Iris and the study procedure for the third, main study. The results of the main survey show a significant preference for Iris. However, disabled students are underrepresented in the sample compared to the general population. We plan to conduct targeted research (e.g., by coordinating with our Disabled Students Program) to address this issue.

References

- [1] Sheryl Burgstahler. 2021. What Higher Education Learned About the Accessibility of Online Opportunities During a Pandemic. *Journal of Higher Education Theory and Practice* 21, 7 (Aug. 2021). doi:10.33423/jhetp.v21i7.4493
- [2] Matthew Butterick. 2010. *Practical Typography*. <https://practicaltypography.com/>
- [3] Tianyuan Cai, Aleena Gertrudes Niklaus, Bernard Kerr, Michael Kraley, and Zoya Bylinskii. 2024. COR Themes for Readability from Iterative Feedback. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–23. doi:10.1145/3613904.3642108
- [4] Centre for Excellence in Universal Design. 2025. The 7 Principles. <https://universaldesign.ie/about-universal-design/the-7-principles>
- [5] Marijn Haverbeke and Adrian Heine. 2025. Prosemirror. <https://prosemirror.net/>
- [6] Aleena Gertrudes Niklaus, Tianyuan Cai, Zoya Bylinskii, and Shaun Wallace. 2023. Digital Reading Rulers: Evaluating Inclusively Designed Rulers for Readers With Dyslexia and Without. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–17. doi:10.1145/3544548.3581367
- [7] Luz Rello and Ricardo Baeza-Yates. 2016. The Effect of Font Type on Screen Readability by People with Dyslexia. *ACM Trans. Access. Comput.* 8, 4 (May 2016), 15:1–15:33. doi:10.1145/2897736
- [8] Martin Schrepp, Andreas Hinderks, and Jörg Thomaschewski. 2017. Design and Evaluation of a Short Version of the User Experience Questionnaire (UEQ-S). *International Journal of Interactive Multimedia and Artificial Intelligence* 4 (Jan. 2017), 103. doi:10.9781/ijimai.2017.09.001
- [9] W3C. 2024. Web Content Accessibility Guidelines 2.2. <https://www.w3.org/TR/WCAG22/>
- [10] WebAIM. 2025. WAVE Web Accessibility Evaluation Tools. <https://wave.webaim.org/>